Loom & Link:
A Real-time Synchronized Scheduler for Long Distance Relationships

Technical Documentation & Architecture Design

Syncing Hearts Across Timezones

Prepared for: Babeș-Bolyai University Academic Review
Author: Tompea Maria Daria
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1 Introduction

1.1 Project Vision

Maintaining a relationship across vast geographic distances introduces a unique set of challenges, primarily revolving around temporal coordination. "Loom & Link" is a Python-based application designed to bridge this gap by providing a visual, interactive, and "cute" scheduling environment. Unlike generic calendar apps, Loom & Link focuses on the emotional and visual connection between two partners.

1.2 Core Objectives

The primary goals of the system are:

- **Real-time Sync**: Every action taken by one partner is instantly visible to the other.
- **Transparent Localization**: Automated timezone conversion that renders the same event correctly on two different local grids.
- **High Aesthetic Appeal**: Using modern UI principles to create a warm, inviting environment.
- **Simplicity**: A code-based pairing system that eliminates complex email-based invites.

2 Design Philosophy

2.1 The "Cute" Aesthetic

The application utilizes a **Glassmorphism** design language. This involves blurred background elements, semi-transparent panels, and soft shadows to create a sense of depth and warmth.

2.1.1 Color Palette

We have curated a 10-color pastel palette for event classification, ensuring that the schedule remains vibrant yet readable:

Color Name	HEX Code	Typical Activity
Strawberry	#FFB7B2	Date Night
Sky Blue	#B2E2F2	Online Gaming
Mint Green	#B2F2BB	Studying Together
Lavender	#D1B2F2	Sleep/Wind Down
Lemon	#FFF1B2	Morning Call

Table 2.1: Subset of the 10-color palette logic.

2.2 Interface Layout

The main interface consists of a weekly grid. The Y-axis represents time in 30-minute increments, while the X-axis represents the days of the week.

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Technical Stack

3.1 Frontend: Python with Flet

The application is built using **Flet**, a framework that enables the creation of Flutter apps using Python. This allows for a reactive UI that behaves like a native application on both desktop and mobile platforms.

3.2 Backend: Supabase Real-time

To facilitate the sharing of tables, **Supabase** is employed. It provides a PostgreSQL database with a real-time listener. When Partner A adds an event, the database emits a broadcast, which Partner B's client receives instantly.

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Timezone Synchronization

4.1 The UTC Standard

To handle users in different regions (e.g., Italy and USA), the application stores all timestamps in the database as **UTC+0**.

4.2 Conversion Algorithm

When the application fetches an event E with a start time T_{utc}, it detects the user's local timezone TZ_{local} and performs the following calculation:

$$T_{local} = T_{utc} + \text{Offset}(TZ_{local}, T_{utc}) \tag{4.1}$$

4.3 Day Boundary Logic

A unique requirement of Loom & Link is handling events that span across different days for different users. For example:

- **Partner A (Rome, UTC+2)**: Adds a "Movie Night" on Sunday at 11:00 PM.
- **Partner B (New York, UTC-4)**: The same event occurs at 5:00 PM on Sunday.

The engine ensures the event appears on the Sunday column for both. However, if the time difference shifts the event to a Monday for one partner, the app intelligently renders it in the Monday column of the *viewer's* local week.

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Application Modules

5.1 Authentication & Pairing

The login system uses a simple alphanumeric code ($6 \leq \text{length} \leq 15$).

```
def validate_code(code):
    if 6 <= len(code) <= 15:
        return True
    return False
```

Once logged in, a user can enter their partner's code to establish a bidirectional link in the `couples` table.

5.2 Event Management

The event popup allows users to input:

- Activity Name (e.g., "Dinner Date").
- Description (e.g., "Eating sushi while on FaceTime").
- Color Selection (One of 10 presets).
- Start and End Time (converted to UTC before saving).

6 Database Schema

6.1 Entity Definitions

6.1.1 User Table

Stores `user_id`, `secret_code`, and `region`.

6.1.2 Event Table

Stores `id`, `couple_id`, `title`, `desc`, `start_time`, `end_time`, and `color_index`.

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Implementation Code

7.1 The Table Component

```
class ScheduleTable(ft.UserControl):
    def build(self):
        # Create 7 columns for the week
        # Populated with local-converted events
        pass
```

7.2 Weekly Reset Logic

At the start of each week (Monday 00:00 UTC), the application triggers a routine to archive the previous week's events and provide an empty table, as per user requirements.

8 Conclusion

Loom & Link provides a robust, aesthetically pleasing solution for long-distance coordination. By abstracting the complexities of timezone mathematics and offering a shared, colorful environment, it fosters closer connection through organized quality time.